

Normalized Cuts and Cheeger's Inequality

Ian Chen

May 31, 2026

1 Introduction

We motivate this discussion through image segmentation: given an array of pixels, we want to segment the image into large, internally-coherent pieces. The GrabCut algorithm [3] assigns a probability z_i of whether pixel i is in the foreground or background (i.e. binary segmentation). To ensure cohesive segments, we formulate the optimization through a graph (typically grid), so that we want to minimize

$$L_{\text{grabcut}}(\mathbf{z}) = \sum_{ij \in E} \left(\underbrace{z_i z_j E_{ij,11}}_{\text{both foreground}} + \underbrace{(1 - z_i) z_j E_{ij,01}}_{\text{boundary}} + \underbrace{z_i (1 - z_j) E_{ij,10}}_{\text{boundary}} + \underbrace{(1 - z_i) (1 - z_j) E_{ij,00}}_{\text{both background}} \right)$$

and can be solved exactly in the case of $E_{ij,00} = 0 = E_{ij,11}$ and $E_{ij,01} = E_{ij,10} > 0$, e.g. via Gaussian kernel,

$$E_{ij,01} = E_{ij,10} = e^{\frac{-1}{2\sigma^2}(\mathbf{f}_i - \mathbf{f}_j)^T(\mathbf{f}_i - \mathbf{f}_j)}$$

where $\mathbf{f}_i$ is a feature associated with the image, such as $(\mathbf{p}, \mathbf{c})$ position and color. More generally, when segmenting into k pieces, we build an affinity matrix and find a cut. In an effort to avoid small trivial segments, the *normalized cut* objective is given as

$$\sigma(S_1, \dots, S_k) = \sum_{\ell=1}^k \frac{w(\partial S_\ell)}{\text{vol}(S_\ell)}$$

where $w(\partial S)$ denotes the sum of weights in the boundary, i.e. all edges with exactly one endpoint in S , and the volume is the sum of weights of all half-edges from S . We penalize small segments (i.e. with little volume), and also segments with poor separation (i.e. with large boundaries). Eigenvalues are natural – they are permutationally covariant – and indeed we will see their use.

2 Laplacians

We start with a graph $G = (V, E)$, with $|V| = n$, $|E| = m$ and non-negative weights $w(\cdot)$, e.g. the grid-graph with Gaussian kernel for images. The edge-incidence matrix $K \in \mathbb{R}^{m \times n}$ indexes rows/cols by edges/nodes, with

$$K_{eu} = \begin{cases} \sqrt{w_e} & e = (u, v), u < v \\ \sqrt{-w_e} & e = (u, v), u > v \\ 0 & \text{otherwise} \end{cases}$$

In particular, for the *characteristic vector* of a set S , $(\mathcal{X}_S)_u = 1_{u \in S}$, then

$$\|K\mathcal{X}_S\|_2^2 = \sum_{uv \in E(S, S)} (\sqrt{w_e} - \sqrt{w_e})^2 + \sum_{uv \in E(V-S, V-S)} (0 - 0)^2 + \sum_{uv \in \partial S} (\pm\sqrt{w_e} - 0)^2 = w(\partial S)$$

K acts as the gradient operator of a vertex-valued function, and K^T acts as the divergence operator of edge-valued vector-field. $K^T K$ is the Laplacian L , and is easy to rewrite as $L = D - A$, for A the adjacency and $D = \text{diag}(A\mathbf{1}_n) = \text{diag}(\mathbf{d})$. The normalized Laplacian is denoted $\tilde{L} = D^{-1/2} L D^{-1/2} = I - D^{-1/2} A D^{-1/2} = I - \tilde{A}$. The Laplacian is positive semi-definite, with $\mathbf{x}^T L \mathbf{x} = \|K\mathbf{x}\|_2^2 = \sum_{ij \in E} w_{ij} (x_i - x_j)^2$.

3 Spectral clustering

Fix $k > 0$. We can re-write the normalized cut objective as a Rayleigh quotient,

$$\sigma(S_1, \dots, S_k) = \sum_{\ell=1}^k \frac{\mathcal{X}_{S_\ell}^T L \mathcal{X}_{S_\ell}}{\mathcal{X}_{S_\ell}^T D \mathcal{X}_{S_\ell}}$$

Solving this problem exactly is NP-Hard, so we use this quadratic form to formulate an abstraction. First, we consider a weighted assignment matrix $H \in \mathbb{R}^{n \times k}$, where $\mathbf{h}_i$, the i -th row indicates the assignment of vertex i :

$$h_{ij} = \begin{cases} 1/\sqrt{\text{vol}(S_j)} & i \in S_j \\ 0 & \text{otherwise} \end{cases}$$

so that $\sigma(S_1, \dots, S_k) = \text{Tr}(H^T L H)$, since $\mathbf{h}_j^T L \mathbf{h}_j = \frac{\mathcal{X}_{S_j}^T L \mathcal{X}_{S_j}}{\text{vol}(S_j)}$. Note also that $\mathbf{h}_j^T D \mathbf{h}_j = \frac{\mathcal{X}_{S_j}^T D \mathcal{X}_{S_j}}{\text{vol}(S_j)} = 1$, so we want to optimize $\text{Tr}(H^T L H)$ subject to $H^T D H = I$ and H has the discrete entries as above. Relaxing discreteness,

$$\hat{H}^{\text{opt}} = \arg \min_{H^T D H = I_k} \text{Tr}(H^T L H) = \arg \min_{J = D^{1/2} H, J^T J = I_k} \text{Tr}(J^T \tilde{L} J) \quad (1)$$

Thus, by a standard exercise in SVD, $\hat{J}^{\text{OPT}}$ has columns of the k -minimum eigenvectors of $\tilde{L}$. A standard rounding is to treat the rows of $\hat{H}^{\text{OPT}}$ as an embedding of the points, and run k -means clustering, which has elegant theory via the Davis-Kahan theorem.

4 Cheeger's Inequality

To finish off Cheeger's inequality, we interpret the above by specializing $k = 2$. We know that $\mathbf{1}_n$ is always in the null-space of L , as $(D - A)\mathbf{1}_n = \mathbf{d} - \mathbf{d} = \mathbf{0}$. Thus, $\mathbf{d}^{1/2}$ is in the null-space of $\tilde{L}$, as $\tilde{L}\mathbf{d}^{1/2} = D^{-1/2} L D^{-1/2} \mathbf{d}^{1/2} = D^{-1/2} L \mathbf{1}_n = \mathbf{0}$. The first column of $\hat{H}^{\text{OPT}}$ is thus $D^{-1/2} \mathbf{d}^{1/2} = \mathbf{1}_n$, and can be ignored (it is independent of G). Indeed, with its 0 contribution, the optimal *relaxed* objective value is the second-smallest eigenvalue of $\tilde{L}$.

Theorem 1. *Let $\lambda_1 \leq \dots \leq \lambda_n$ be the eigenvalues of $\tilde{L}$, $\sigma(G) = \min_{S \subset V} \sigma(S, V - S)$. Then, $\lambda_2 \leq \sigma(G)$.*

Now, let us address the problem of rounding the solution. We proceed via sweep-cuts:

- Compute $\mathbf{x}$ 2nd eigenvector of the normalized Laplacian $D^{-1/2} L D^{-1/2} = \tilde{L}$.
- Compute $\{S_\tau = \{v \in V : \mathbf{x}_2(v) \leq \tau\}\}$, and take the minimum conductance among all τ .

WLOG, assume that $\mathbf{x}$ is ascending and $x_1^2 + x_n^2 = 1$. Suppose we sample τ according to $2|t|$ (so that we have squared mass), and we show that $\Pr_\tau(\sigma(S_\tau) \leq \sqrt{2\lambda_2}) > 0$. Indeed, it suffices that¹

$$\mathbf{E}(\min(\text{vol}(S), \text{vol}(\bar{S}))) = \mathbf{x}^T D \mathbf{x}, \quad \mathbf{E}(|\partial S|) \leq \sqrt{\mathbf{x}^T \tilde{L} \mathbf{x}} \sqrt{2\mathbf{x}^T D \mathbf{x}} = \sqrt{2\lambda_2} \mathbf{x}^T D \mathbf{x}$$

¹For $Y \neq 0$, then $\Pr\left(\frac{X}{Y} \leq \frac{\mathbf{E}(X)}{\mathbf{E}(Y)}\right) > 0$, since $Z = X - \mathbf{E}(X)Y/\mathbf{E}(Y)$ is zero everywhere, else has positive probability of both signs.

For the first bound, for both signs of $x(u)$, indeed (as the minimum is determined by the 0 point, since $\mathbf{x} \perp \mathbf{d}$)

$$\begin{aligned} \Pr(u \in \arg \min(\text{vol}(S), \text{vol}(\bar{S}))) &= \begin{cases} \tau \in [x(u), 0] = x(u)^2 & x(u) < 0 \\ \tau \in [0, x(u)] = x(u)^2 & x(u) > 0 \end{cases} \\ \implies \mathbf{E}(\min(\text{vol}(S), \text{vol}(\bar{S}))) &= \sum_u \deg(u) x(u)^2 = \mathbf{x}^T D \mathbf{x} \end{aligned}$$

For the second, we apply Cauchy-Schwartz to connect ℓ_1 and ℓ_2 norms,

$$\begin{aligned} \Pr(uv \in E(S, \bar{S})) &= \begin{cases} |x(u)^2 - x(v)^2| & \text{different signs} \\ x(u)^2 + x(v)^2 & \text{same signs} \end{cases} \leq |x(u) - x(v)|(|x(u)| + |x(v)|) \\ \implies \mathbf{E}(|\partial S|) &\leq \sum_{uv \in E} |x(u) - x(v)|(|x(u)| + |x(v)|) \\ &\leq \sqrt{\sum_{uv \in E} w(uv)(x(u) - x(v))^2} \sqrt{\sum_{uv \in E} w(uv)(|x(u)| + |x(v)|)^2} \leq \sqrt{\mathbf{x}^T L \mathbf{x}} \sqrt{2\mathbf{x}^T D \mathbf{x}} \end{aligned}$$

Theorem 2. Let $\lambda_1 \leq \dots \leq \lambda_n$ be the eigenvalues of $\tilde{L}$, $\sigma(G) = \min_{S \subset V} \sigma(S, V - S)$. Then, $\sigma(G) \leq 2\sqrt{2\lambda_2}$.

Putting this together, we find Cheeger's inequality, typically stated in terms of the conductance

$$\varphi(G) = \min_{S \subset V} \frac{w(\partial S)}{\min(\text{vol}(S), \text{vol}(V - S))} \quad \frac{\lambda_2}{2} \leq \varphi(G) \leq \sqrt{2\lambda_2}$$

5 Bibliographic Notes

The background of image segmentation is adapted from Professor Forsyth's unreleased textbook²[1]. The section of Laplacians is mostly standard but is adapted from the Fall 2009 instance of MIT 18-409³. The spectral cuts is adapted from a tutorial in Luxburg [2], as well as Daniel Spielman's book on spectral graph theory. The proof of Cheeger's inequality is described in Trevisan's 2011 offering of CS359⁴.

References

- [1] David Forsyth. *Computer Vision*. TBD, 2026.
- [2] Ulrike von Luxburg. "A tutorial on spectral clustering". In: *Statistics and Computing* 17.4 (Aug. 2007), pp. 395–416. ISSN: 1573-1375. DOI: 10.1007/s11222-007-9033-z. URL: <http://dx.doi.org/10.1007/s11222-007-9033-z>.
- [3] Carsten Rother, Vladimir Kolmogorov, and Andrew Blake. "GrabCut: interactive foreground extraction using iterated graph cuts". In: *ACM Transactions on Graphics* 23.3 (Aug. 2004), pp. 309–314. ISSN: 1557-7368. DOI: 10.1145/1015706.1015720. URL: <http://dx.doi.org/10.1145/1015706.1015720>.

²Draft at Spring 2026 CS 543 offering, <http://luthuli.cs.uiuc.edu/~daf/Courses/CV2026/classwebpage.html>

³<https://ocw.mit.edu/courses/18-409-topics-in-theoretical-computer-science-an-algorithmists-toolkit-fall-2009/>

⁴<https://theory.stanford.edu/~trevisan/cs359g/>